

MATH 316 — Review Topics for Exam 2 (Homework 4-5)
Fall 2026

2nd order linear, constant coeff, ODEs $ax'' + bx' + cx = f(t)$

Self-assessment

- I can find the general solution to the homogeneous equation $ax'' + bx' + cx = 0$
- For the homogeneous equation $ax'' + bx' + cx = 0$ I can
 - find the general solution
 - Solve an initial value problem
 - Plot the solution in each case ($b^2 - 4ac > 0$, $b^2 - 4ac < 0$, $b^2 - 4ac = 0$)
- For the nonhomogeneous equation $ax'' + bx' + cx = f(t)$ for special right hand sides (eg, polynomials, exponentials, sine/cosine) I can
 - State the form of the general solution
 - Find the general solution, by properly finding the “undetermined coefficients”
 - Solve an initial value problem
- I can state a simple linear model differential equation for a spring and interpret the meaning of each parameter.
- I can relate the mathematical solution to the behaviour of the physical system.
- I can explain any of the terms *beats*, *resonance*, *forced/steady state solution*, *damping*, *overdamped*, *underdamped*, *critically damped* to my ... (brother/sister, friend, mom/dad, cat, dog).

1. Homogeneous case $ax'' + bx' + cx = 0$

- For constant coefficient case: Look for $x = e^{rt} \Rightarrow$ characteristic equation
- Solve the homogeneous initial value problem

$$ax'' + bx' + cx = 0, \quad x(t_0) = x_0, \quad x'(t_0) = x'_0$$

in any of the 3 cases: (1) real roots of characteristic equation, (2) complex roots, (3) repeated real roots.

- Be able to sketch graphs of solutions that captures the general behaviour and the limiting behaviour as $t \rightarrow \pm\infty$
- Understand the dependence of the solution on parameters.

2. Nonhomogeneous case: $ax'' + bx' + cx = f(t)$

Solution is always of the form $x(t) = x_h(t) + x_p(t)$ where x_h is the general solution to the homogeneous problem and x_p is a particular solution. Use either

- Undetermined coefficients: Guess Ansatz depending on $f(t)$ and x_h (constant coefficient case)

- If asked for, only state the general form of the solution. Example: $x = c_1 e^t + c_2 e^{-2t} + A \cos t + B \sin t$

If asked for, state the general solution. Example: $x = c_1 e^t + c_2 e^{-2t} + 2 \cos t - \frac{1}{2} \sin t$

If asked for, find the solution of an initial value problem. Example: $x = 3e^t - 2e^{-2t} + 2 \cos t - \frac{1}{2} \sin t$

3. Applications

- Understand the solutions of the harmonic oscillator

$$mx'' + \gamma x' + kx = f(t), \quad x(t_0) = x_0, \quad x'(t_0) = x'_0$$

where all coefficients are positive $m, \gamma, k > 0$, for each of

- unforced case: $f(t) = 0$
 - * undamped ($\gamma = 0$)
 - * underdamped ($0 < \gamma < \gamma_c$, what is γ_c ?)
 - * critically damped ($\gamma = \gamma_c$)
 - * overdamped ($\gamma > \gamma_c$)
- forced case: $f(t) \neq 0$
 - * undamped, periodic forcing: beats and resonance
 - * damped, periodic forcing: what happens to beats and resonance if you add damping?
 - * find the transient and the forced (or steady state) solution.
 - * find period, natural period, quasi-periods.

That is, in each case know what to expect as well as how to find the solution.

- Solve equations with parameters ($m, \gamma, k, \delta, \nu$, etc) in them.

4. Understanding trig functions

- Write the sum of a sine and a cosine function with equal period as a shifted cosine (or sine)

$$A \sin(\omega t) + B \cos(\omega t) = R \cos(\omega t - \varphi)$$

where $R = \sqrt{A^2 + B^2}$, $R \sin \varphi = A$, $R \cos \varphi = B$.

- Write sums of sines and cosines with unequal periods as a product of sines or cosines, using the following relations

$$\cos(a) + \cos(b) = 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

$$\cos(a) - \cos(b) = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$\sin(a) + \sin(b) = 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right)$$

(If needed, these relations will be given to you in the exam, no need to memorize them.)

- Use these results to plot, understand connection to beats.

5. Complex variables and Euler's Formula

- State Euler's Formula
- Write complex numbers in Cartesian form or in polar form